

Matthew Tsui

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Skills

Programming Languages

- Java
- C++
- Python

Applications

- Eclipse
- Visual Studio Code

O/S

- Windows XP/7/10/11

Projects

TestMaker

- Applied object-oriented design practices to code by creating sub-classes and setting data members as private or protected, achieving effective encapsulation so that only authorized classes could access the data
- Utilized Junit Test to white box test for bugs by checking each method for extreme cases and many different cases ensuring that the code worked as intended
- Created multiple classes to create a quiz from questions of certain difficulty levels or all questions provided by the user
- Developed a robust and efficient object-oriented programming approach by creating child classes that inherited essential data and methods from parent classes, resulting in significant code reuse and a streamlined development process

MatrixDriver

- Designed a matrix class to store and manipulate 2D vectors of integers by using methods to perform calculations on the vectors
- Overloaded operators to perform different calculations on the vectors and print them out
- Diligently conducted extensive research on overloading 2D vectors online, leading to a deep understanding of pointers
- Efficiently organized free time in between school and work by working on code for an hour at a time and then taking a 30-minute break, resulting in increased productivity and less burnout

Maze

- Developed a sophisticated algorithm using C++ that effectively navigated a matrix of integers to find a zero-sum path through recursive programming
- Successfully utilized a 2D vector to store and display the path taken through the maze, to provide a more precise visualization of the solution
- Utilized file input to read data from a text file to initialize a 2D array, providing a maze for the program to navigate through

Guess Number Game

- Successfully coded a game where the user is given several attempts to guess a random number in a specific range by using things I learned in class, such as loops and the scanner class resulting in a fun game
- Diligently pursued a solution for errors in code by searching the internet, resulting in the learning of try and catch outside of the classroom setting
- Utilized practices of Software Development Life Cycle and top-down design by planning out the code and testing it for errors, resulting in an easier time in producing the program

Education

Simon Fraser University, Burnaby, BC
Bachelor's of Applied Science in Computer Science

September 2024 - Present

Employment History

T&T Supermarket, Burnaby, BC
Cashier

December 2016 – Present

- Successfully decorated the store within the deadline by planning out a schedule for hanging the decorations in each department, ensuring a positive experience for customers and employees
- Logically alleviated 3 to 4 customer complaints per week by remaining calm even when faced with a rude or difficult customer and following company policies to ensure customer satisfaction
- Partnered with another co-worker in closing the store by dividing work fairly so that each person knew their responsibilities and helped each other when needed, reducing the closing time by 50%
- Clearly explained to up to 15 new employees how to do daily tasks utilizing a combination of demonstration, verbal explanations, and a training checklist, ensuring the employee meets company expectations